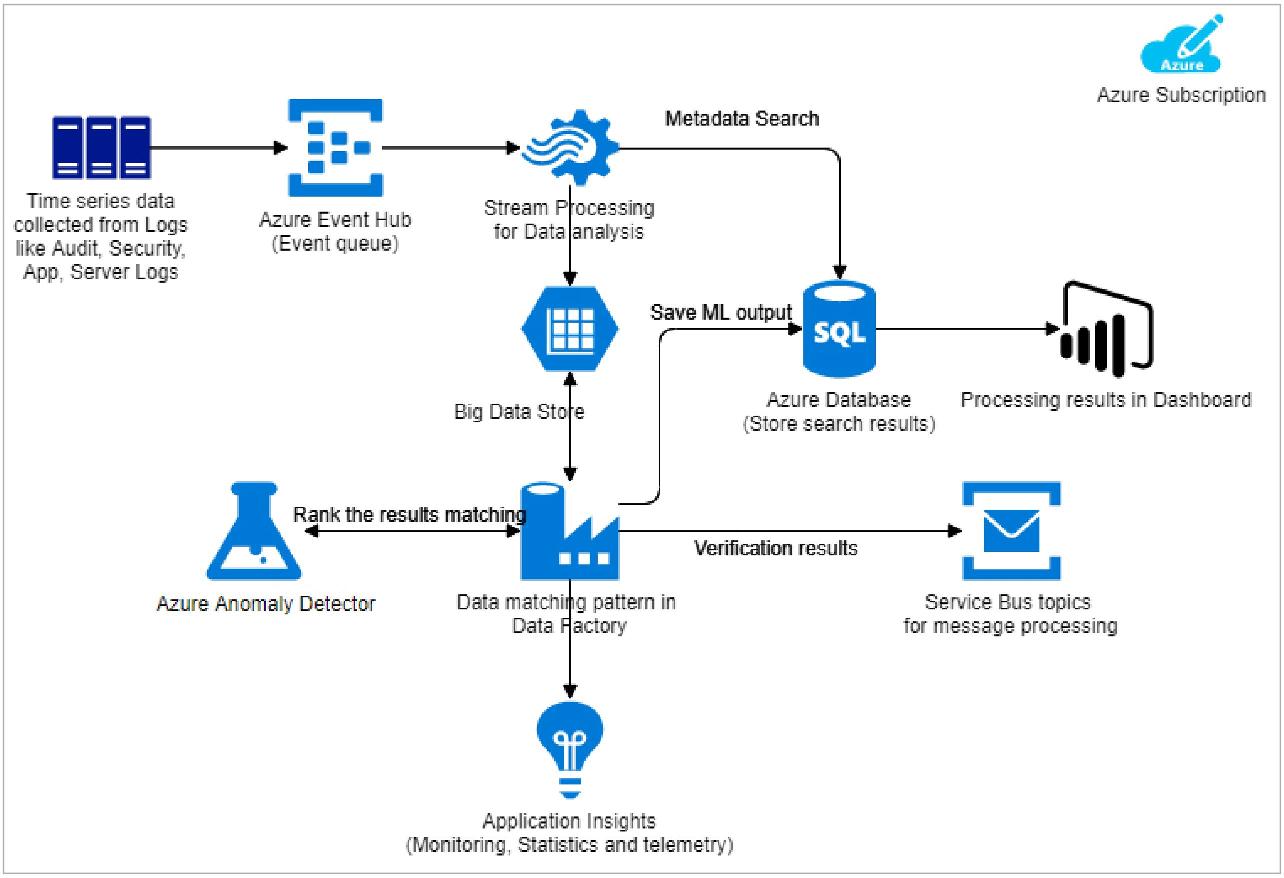

Figure 5: Azure Anomaly Detector architecture

speech recognition, emotion detection, and speech and language based advanced features can be added to applications.

These are available as REST APIs, client library SDKs and user interfaces. Azure Cognitive Services have tons of useful benefits for developers like:

a. Easy integration of pretrained and customised AI models in applications
b. Strong deployment of cognitive services from cloud to edge
c. Improved customer experience with simple and intuitive GUI in applications

Azure Form Recognizer

Azure has introduced a new service called Form Recognizer, which is built on cognitive services and OCR methods. It enables character extraction from pre-printed forms like invoices and bills. Currently, it works for printed character extraction and can be used to extract text from PDFs, office documents, and images. Feature enhancement is in progress for recognition of handwritten forms/invoices/bills.

Azure ML Studio based OCR APIs is a supervised approach, where training and manual correction is required to improve the recognition rate of extracted text. Forms Recognizer is an unsupervised model, which improves the overall efficiency of the text recognition process.

Azure anomaly detection

Anomaly detection is a Big Data processing solution that aims at finding rare event patterns from a collection of patterns by creating pattern groups based on occurrences. For example, it can identify unusual login behaviour from a collection of audit logs, or unusual server failures from a server log.

In an enterprise, these logs are ever growing in size and have various patterns. Identifying which of these are valid events and which are suspicious ones is a big task.

Azure provides a sophisticated AI based service called Azure Anomaly Detector which uses API calls to use time series data like logs, enabling AI based detection of unusual event patterns. Before Azure introduced the Anomaly Detector, the Azure Machine Learning service was used to identify patterns of events, and use these for training the system to identify unusual behaviours.

With the introduction of anomaly detection, the system can self-train to build the knowledge base of patterns